

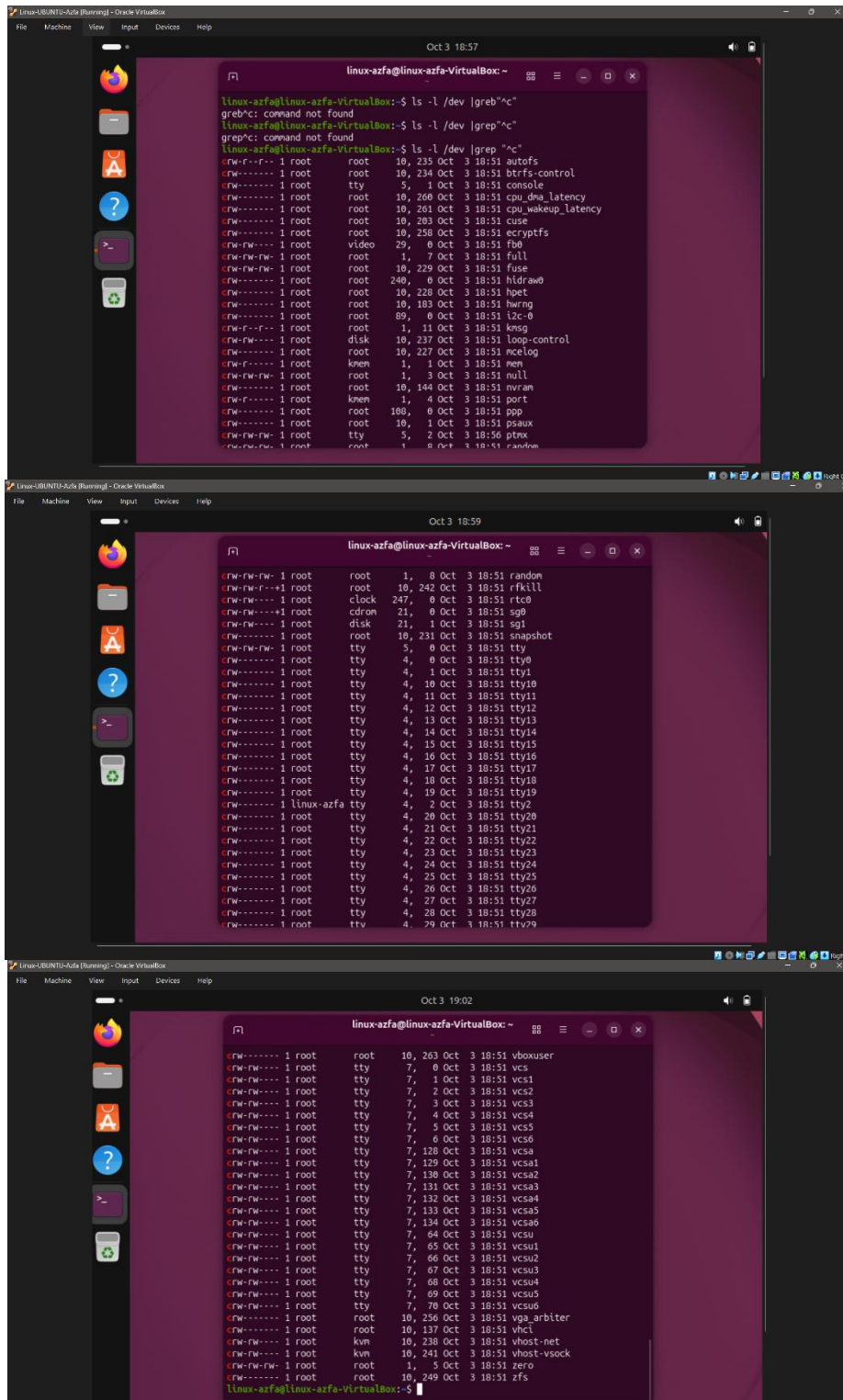
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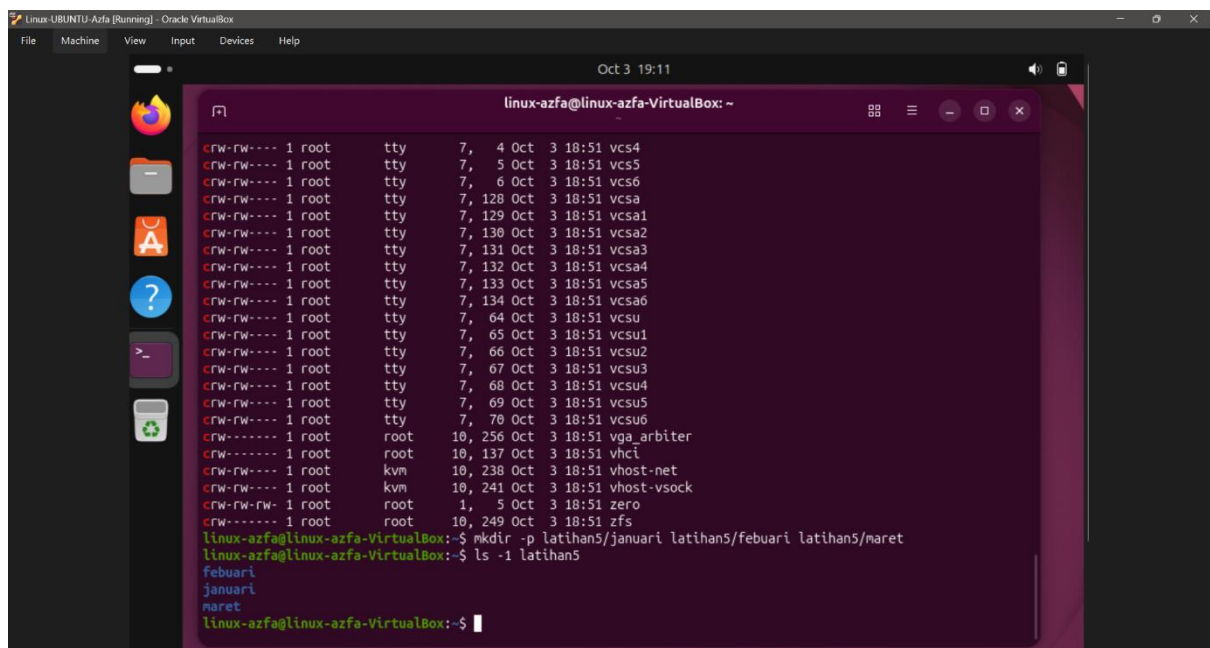
## Tugas 5 Sistem Operasi

1. Lihat peralatan I/O, character device, yang ada pada sistem komputer.



```
linux-azfa@linux-azfa-VirtualBox: ~  
linux-azfa@linux-azfa-VirtualBox:~$ ls -l /dev | grep '^c$'  
linux-azfa@linux-azfa-VirtualBox:~$ ls -l /dev | grep '^c$'  
crw-rw-rw- 1 root root 10, 235 Oct 3 18:51 autofs  
crw-rw-rw- 1 root root 10, 234 Oct 3 18:51 btrfs-control  
crw-rw-rw- 1 root tty 5, 1 Oct 3 18:51 console  
crw-rw-rw- 1 root root 10, 268 Oct 3 18:51 cpu_dma_latency  
crw-rw-rw- 1 root root 10, 261 Oct 3 18:51 cpu_wakeup_latency  
crw-rw-rw- 1 root root 10, 263 Oct 3 18:51 cuse  
crw-rw-rw- 1 root root 10, 258 Oct 3 18:51 ecryptfs  
crw-rw-rw- 1 root video 29, 0 Oct 3 18:51 fb0  
crw-rw-rw- 1 root root 1, 7 Oct 3 18:51 full  
crw-rw-rw- 1 root root 10, 229 Oct 3 18:51 fuse  
crw-rw-rw- 1 root root 240, 0 Oct 3 18:51 hidraw0  
crw-rw-rw- 1 root root 10, 228 Oct 3 18:51 hpet  
crw-rw-rw- 1 root root 10, 183 Oct 3 18:51 hwrng  
crw-rw-rw- 1 root root 89, 0 Oct 3 18:51 i2c-0  
crw-rw-rw- 1 root root 1, 11 Oct 3 18:51 klogd  
crw-rw-rw- 1 root disk 10, 237 Oct 3 18:51 loop-control  
crw-rw-rw- 1 root root 10, 227 Oct 3 18:51 mcelog  
crw-rw-rw- 1 root knmem 1, 1 Oct 3 18:51 mem  
crw-rw-rw- 1 root root 1, 3 Oct 3 18:51 null  
crw-rw-rw- 1 root root 10, 144 Oct 3 18:51 nvram  
crw-rw-rw- 1 root knmem 1, 4 Oct 3 18:51 port  
crw-rw-rw- 1 root root 188, 0 Oct 3 18:51 ppp  
crw-rw-rw- 1 root root 10, 1 Oct 3 18:51 psaux  
crw-rw-rw- 1 root tty 5, 2 Oct 3 18:56 ptmx  
crw-rw-rw- 1 root root 1, 8 Oct 3 18:51 random  
crw-rw-rw- 1 root root 1, 8 Oct 3 18:51 random  
crw-rw-rw- 1 root root 10, 242 Oct 3 18:51 rftkill  
crw-rw-rw- 1 root clock 247, 0 Oct 3 18:51 rtc0  
crw-rw-rw- 1 root cdrom 21, 0 Oct 3 18:51 sg0  
crw-rw-rw- 1 root disk 21, 1 Oct 3 18:51 sg1  
crw-rw-rw- 1 root root 10, 231 Oct 3 18:51 snapshot  
crw-rw-rw- 1 root tty 5, 0 Oct 3 18:51 tty  
crw-rw-rw- 1 root tty 4, 0 Oct 3 18:51 tty0  
crw-rw-rw- 1 root tty 4, 1 Oct 3 18:51 tty1  
crw-rw-rw- 1 root tty 4, 10 Oct 3 18:51 tty10  
crw-rw-rw- 1 root tty 4, 11 Oct 3 18:51 tty11  
crw-rw-rw- 1 root tty 4, 12 Oct 3 18:51 tty12  
crw-rw-rw- 1 root tty 4, 13 Oct 3 18:51 tty13  
crw-rw-rw- 1 root tty 4, 14 Oct 3 18:51 tty14  
crw-rw-rw- 1 root tty 4, 15 Oct 3 18:51 tty15  
crw-rw-rw- 1 root tty 4, 16 Oct 3 18:51 tty16  
crw-rw-rw- 1 root tty 4, 17 Oct 3 18:51 tty17  
crw-rw-rw- 1 root tty 4, 18 Oct 3 18:51 tty18  
crw-rw-rw- 1 root tty 4, 19 Oct 3 18:51 tty19  
crw-rw-rw- 1 linux-azfa tty 4, 2 Oct 3 18:51 tty2  
crw-rw-rw- 1 root tty 4, 20 Oct 3 18:51 tty20  
crw-rw-rw- 1 root tty 4, 21 Oct 3 18:51 tty21  
crw-rw-rw- 1 root tty 4, 22 Oct 3 18:51 tty22  
crw-rw-rw- 1 root tty 4, 23 Oct 3 18:51 tty23  
crw-rw-rw- 1 root tty 4, 24 Oct 3 18:51 tty24  
crw-rw-rw- 1 root tty 4, 25 Oct 3 18:51 tty25  
crw-rw-rw- 1 root tty 4, 26 Oct 3 18:51 tty26  
crw-rw-rw- 1 root tty 4, 27 Oct 3 18:51 tty27  
crw-rw-rw- 1 root tty 4, 28 Oct 3 18:51 tty28  
crw-rw-rw- 1 root tty 4, 29 Oct 3 18:51 tty29  
crw-rw-rw- 1 root root 10, 263 Oct 3 18:51 vboxuser  
crw-rw-rw- 1 root tty 7, 0 Oct 3 18:51 vcs  
crw-rw-rw- 1 root tty 7, 1 Oct 3 18:51 vcs1  
crw-rw-rw- 1 root tty 7, 2 Oct 3 18:51 vcs2  
crw-rw-rw- 1 root tty 7, 3 Oct 3 18:51 vcs3  
crw-rw-rw- 1 root tty 7, 4 Oct 3 18:51 vcs4  
crw-rw-rw- 1 root tty 7, 5 Oct 3 18:51 vcs5  
crw-rw-rw- 1 root tty 7, 6 Oct 3 18:51 vcs6  
crw-rw-rw- 1 root tty 7, 128 Oct 3 18:51 vcsa  
crw-rw-rw- 1 root tty 7, 129 Oct 3 18:51 vcsa1  
crw-rw-rw- 1 root tty 7, 130 Oct 3 18:51 vcsa2  
crw-rw-rw- 1 root tty 7, 131 Oct 3 18:51 vcsa3  
crw-rw-rw- 1 root tty 7, 132 Oct 3 18:51 vcsa4  
crw-rw-rw- 1 root tty 7, 133 Oct 3 18:51 vcsa5  
crw-rw-rw- 1 root tty 7, 134 Oct 3 18:51 vcsa6  
crw-rw-rw- 1 root tty 7, 64 Oct 3 18:51 vcsu  
crw-rw-rw- 1 root tty 7, 65 Oct 3 18:51 vcsu1  
crw-rw-rw- 1 root tty 7, 66 Oct 3 18:51 vcsu2  
crw-rw-rw- 1 root tty 7, 67 Oct 3 18:51 vcsu3  
crw-rw-rw- 1 root tty 7, 68 Oct 3 18:51 vcsu4  
crw-rw-rw- 1 root tty 7, 69 Oct 3 18:51 vcsu5  
crw-rw-rw- 1 root tty 7, 70 Oct 3 18:51 vcsu6  
crw-rw-rw- 1 root root 10, 256 Oct 3 18:51 vga_arbiter  
crw-rw-rw- 1 root root 10, 137 Oct 3 18:51 vhost  
crw-rw-rw- 1 root kvm 10, 238 Oct 3 18:51 vhost-net  
crw-rw-rw- 1 root kvm 10, 241 Oct 3 18:51 vhost-vsock  
crw-rw-rw- 1 root root 1, 5 Oct 3 18:51 zero  
crw-rw-rw- 1 root root 10, 249 Oct 3 18:51 zfs  
linux-azfa@linux-azfa-VirtualBox:~$
```

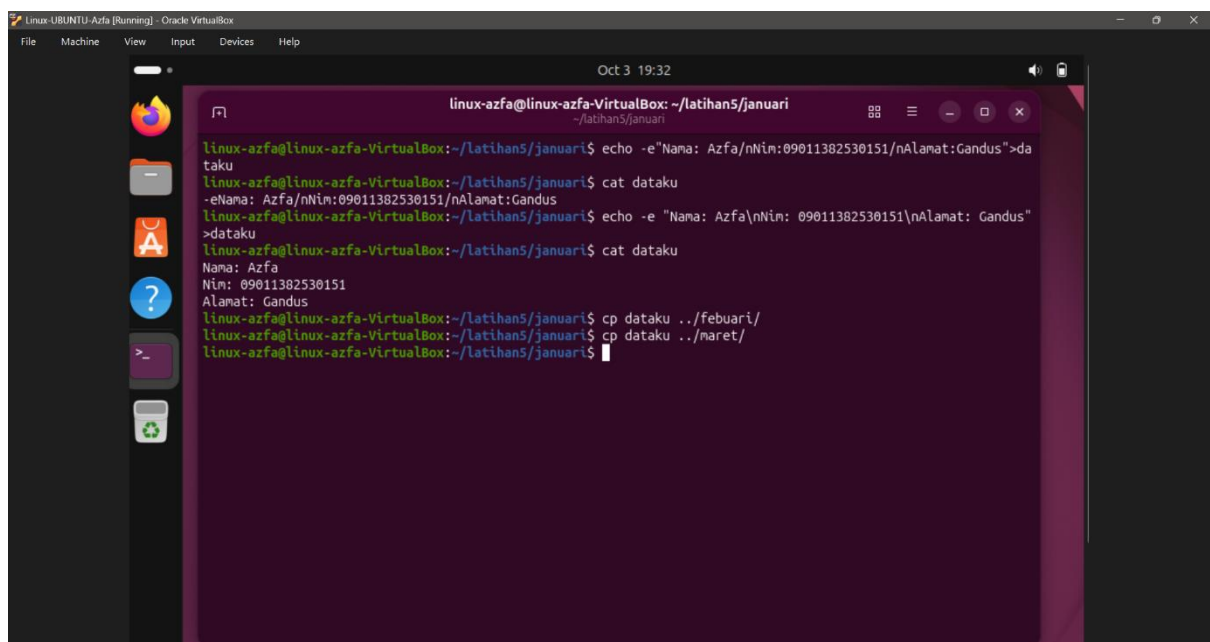
2. Buatlah sub direktori januari, februari dan maret sekaligus pada direktori latihan 5.



The screenshot shows a terminal window titled 'linux-azfa@linux-azfa-VirtualBox: ~'. The terminal displays a list of system files and directories, including 'vcs4' through 'zfs'. Below this list, the user enters the command 'mkdir -p latihan5/januari latihan5/februari latihan5/maret', followed by 'ls -l latihan5', which shows the newly created subdirectories 'februari', 'januari', and 'maret'.

```
linux-azfa@linux-azfa-VirtualBox: ~  
crw-rw---- 1 root tty 7, 4 Oct 3 18:51 vcs4  
crw-rw---- 1 root tty 7, 5 Oct 3 18:51 vcs5  
crw-rw---- 1 root tty 7, 6 Oct 3 18:51 vcs6  
crw-rw---- 1 root tty 7, 128 Oct 3 18:51 vcsa  
crw-rw---- 1 root tty 7, 129 Oct 3 18:51 vcsa1  
crw-rw---- 1 root tty 7, 130 Oct 3 18:51 vcsa2  
crw-rw---- 1 root tty 7, 131 Oct 3 18:51 vcsa3  
crw-rw---- 1 root tty 7, 132 Oct 3 18:51 vcsa4  
crw-rw---- 1 root tty 7, 133 Oct 3 18:51 vcsa5  
crw-rw---- 1 root tty 7, 134 Oct 3 18:51 vcsa6  
crw-rw---- 1 root tty 7, 64 Oct 3 18:51 vcsu  
crw-rw---- 1 root tty 7, 65 Oct 3 18:51 vcsu1  
crw-rw---- 1 root tty 7, 66 Oct 3 18:51 vcsu2  
crw-rw---- 1 root tty 7, 67 Oct 3 18:51 vcsu3  
crw-rw---- 1 root tty 7, 68 Oct 3 18:51 vcsu4  
crw-rw---- 1 root tty 7, 69 Oct 3 18:51 vcsu5  
crw-rw---- 1 root tty 7, 70 Oct 3 18:51 vcsu6  
crw-rw---- 1 root root 10, 256 Oct 3 18:51 vga_arbiter  
crw-rw---- 1 root root 10, 137 Oct 3 18:51 vhci  
crw-rw---- 1 root kvm 10, 238 Oct 3 18:51 vhost-net  
crw-rw---- 1 root kvm 10, 241 Oct 3 18:51 vhost-vsock  
crw-rw-rw- 1 root root 1, 5 Oct 3 18:51 zero  
crw-rw---- 1 root root 10, 249 Oct 3 18:51 zfs  
linux-azfa@linux-azfa-VirtualBox:~$ mkdir -p latihan5/januari latihan5/februari latihan5/maret  
linux-azfa@linux-azfa-VirtualBox:~$ ls -l latihan5  
februari  
januari  
maret  
linux-azfa@linux-azfa-VirtualBox:~$
```

3. Buatlah file dataku yang berisi nama, nim dan alamat anda pada sub direktori januari dan copy-kan file tersebut ke sub direktori februari dan maret.



The screenshot shows a terminal window titled 'linux-azfa@linux-azfa-VirtualBox: ~/latihan5/januari'. The user creates a file named 'dataku' with the command 'echo -e "Nama: Azfa\nNim:09011382530151\nAlamat:Gandus">dataku'. They then verify the file's contents with 'cat dataku'. Finally, they copy the file to the 'februari' and 'maret' subdirectories using 'cp dataku ../februari/' and 'cp dataku ../maret/'.

```
linux-azfa@linux-azfa-VirtualBox: ~/latihan5/januari  
linux-azfa@linux-azfa-VirtualBox:~/latihan5/januari$ echo -e "Nama: Azfa\nNim:09011382530151\nAlamat:Gandus">dataku  
linux-azfa@linux-azfa-VirtualBox:~/latihan5/januari$ cat dataku  
-eNama: Azfa\nNim:09011382530151\nAlamat:Gandus  
linux-azfa@linux-azfa-VirtualBox:~/latihan5/januari$ echo -e "Nama: Azfa\nNim: 09011382530151\nAlamat: Gandus">dataku  
linux-azfa@linux-azfa-VirtualBox:~/latihan5/januari$ cat dataku  
Nama: Azfa  
Nim: 09011382530151  
Alamat: Gandus  
linux-azfa@linux-azfa-VirtualBox:~/latihan5/januari$ cp dataku ../februari/  
linux-azfa@linux-azfa-VirtualBox:~/latihan5/januari$ cp dataku ../maret/  
linux-azfa@linux-azfa-VirtualBox:~/latihan5/januari$
```

4. Ubahlah ijin akses file dataku pada sub direktori januari sehingga group dan others dapat melakukan write.

```
linux-azfa@linux-azfa-VirtualBox:~/latihan5/januari$ chmod go+w dataku
linux-azfa@linux-azfa-VirtualBox:~/latihan5/januari$ chmod 755 ../februari/dataku
linux-azfa@linux-azfa-VirtualBox:~/latihan5/januari$ chmod 777 ../maret/dataku
linux-azfa@linux-azfa-VirtualBox:~/latihan5/januari$
```

5. Ubahlah ijin akses file dataku pada sub direktori februari sehingga user dapat melakukan baik write, read maupun execute, tetapi group dan others hanya bisa read dan execute.

```
linux-azfa@linux-azfa-VirtualBox:~/latihan5/januari$ chmod go+w dataku
linux-azfa@linux-azfa-VirtualBox:~/latihan5/januari$ chmod 755 ../februari/dataku
linux-azfa@linux-azfa-VirtualBox:~/latihan5/januari$ chmod 777 ../maret/dataku
linux-azfa@linux-azfa-VirtualBox:~/latihan5/januari$
```

6. Ubahlah ijin akses file dataku pada sub direktori maret sehingga semua dapat melakukan write, read dan execute.

```
linux-azfa@linux-azfa-VirtualBox:~/latihan5/januari$ chmod go+w dataku
linux-azfa@linux-azfa-VirtualBox:~/latihan5/januari$ chmod 755 ../februari/dataku
linux-azfa@linux-azfa-VirtualBox:~/latihan5/januari$ chmod 777 ../maret/dataku
linux-azfa@linux-azfa-VirtualBox:~/latihan5/januari$
```

7. Hapuslah direktori maret.

```
linux-azfa@linux-azfa-VirtualBox:~/latihan5/januari$ rm -rf ../maret
linux-azfa@linux-azfa-VirtualBox:~/latihan5/januari$ rm-rf../maret
bash: rm-rf../maret: No such file or directory
linux-azfa@linux-azfa-VirtualBox:~/latihan5/januari$ ls -l ../
februari
januari
linux-azfa@linux-azfa-VirtualBox:~/latihan5/januari$
```

8. Ubahkan kepemilikan sub direktori februari sehingga user dan group hanya dapat melakukan read, dan cobalah untuk membuat direktori baru haha pada sub direktori februari.

```
linux-azfa@linux-azfa-VirtualBox:~/latihan5/januari$ chmod 444 ../februari
linux-azfa@linux-azfa-VirtualBox:~/latihan5/januari$ chmod 444../februari
chmod: missing operand
Try 'chmod --help' for more information.
linux-azfa@linux-azfa-VirtualBox:~/latihan5/januari$ mkdir../februari/haha
bash: mkdir../februari/haha: No such file or directory
linux-azfa@linux-azfa-VirtualBox:~/latihan5/januari$ mkdir ../februari/haha
mkdir: Permission denied
linux-azfa@linux-azfa-VirtualBox:~/latihan5/januari$
```

9. Modifikasi umask dari file dataku pada sub direktori januari menjadi 027 dan berapakan nilai default-nya?

```
linux-azfa@linux-azfa-VirtualBox:~/latihan5/januari$ unmask 027
Command 'unmask' not found, did you mean:
  command 'unmass' from deb unmass (0.9-8build1)
Try: sudo apt install <deb name>
linux-azfa@linux-azfa-VirtualBox:~/latihan5/januari$ umask 027
linux-azfa@linux-azfa-VirtualBox:~/latihan5/januari$
```

10. Buatlah link dari file dataku ke file dataku.ini dan file dataku.juga dan dengan perintah list perhatikan berapa link yang terjadi?

```
linux-azfa@linux-azfa-VirtualBox:~/latihan5/januari$ ln dataku dataku.ini
linux-azfa@linux-azfa-VirtualBox:~/latihan5/januari$ ln dataku dataku.juga
linux-azfa@linux-azfa-VirtualBox:~/latihan5/januari$ ls -l
dataku
dataku.ini
dataku.juga
linux-azfa@linux-azfa-VirtualBox:~/latihan5/januari$
```